WIL_Task 2

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CENTRAL KAROO DISTRICT COMMUNITY NETWORK: TECHNICAL SPECIFICATIONS AND IMPLEMENTATION REPORT

1. HARDWARE REQUIREMENTS

Gido et al. (2018), In building a wireless community network within the Central Karoo District, we will need hardware that will support such a project in a rural setting. The hardware infrastructure needs to be able to cover long distances, wide range of temperature differences, dust, lightning and power problems. Our proposed hardware will ensure connectivity of 3,000 households, 3 district hospitals, and 28 municipal offices.

Our hardware infrastructure will include core routers, wireless backhaul radios, distribution radios, WiFi access points and solar power installations at some remote tower sites.

1.1 Core and Distribution Hardware

1.1.1 Central Gateway and Core Router: MikroTik CCR2116-12G-4S+

The MikroTik CCR2116-12G-4S+ is recommended as the core router at the Network Operations Centre (NOC) in Beaufort West.

Specs: Sixteen-core 64-bit ARM CPU running at 2.0 GHz, 16 GB RAM, 13 Gigabit Ethernet ports and four 10G SFP+ ports. The device has two AC power supplies.

Network role: The router handles traffic between municipal offices, hospitals, wireless distribution points, and community access networks. The router performs Layer 3 routing, such as OSPF and BGP, VLAN segmentation, firewall policies, bandwidth management, and RADIUS authentication services integration.

Redundancy: Two core routers will increase the network's reliability. The proper failover setup and online UPS will help ensure that the service remains available during equipment and electrical power outages.

The second router needs to be set up in such a way that it takes over when the first one fails. Using two routers does not make the network redundant per se; failover is needed as well.

1.1.2 High-Capacity Backhaul Radio Units: Ubiquiti airFiber 11 (AF-11)

This product is recommended for wireless backhaul links between major network sites over long distances.

Specifications: It operates within the frequency range of 10.7 – 11.7 GHz and is capable of offering manufacturer-guaranteed throughput of over 1.2 Gbps under optimal configurations.

Functionality within the network: These radios will set up high-capacity wireless links between certain network sites, possibly between Beaufort West and other towns or intermediate relay towers.

Advantages: Utilizing a suitable licensed spectrum can decrease interference from the unlicensed spectrum (2.4 GHz and 5 GHz). However, it should be noted that license for the spectrum will need to be acquired. Also, performance of these devices depends on the line of sight, link distance, antenna alignment and channel configuration.

Radio path survey should be performed prior to installation of these devices in order to ensure their capabilities meet requirements.

1.1.3 Backhaul of Medium Range Between Sites: MikroTik NetMetal 5 and mANTBox

Equipment: MikroTik NetMetal 5 and mANTBox will be used for connectivity distribution between backbone locations and secondary repeaters as well as local communities.

Specifications: Equipment chosen has support for 5 GHz wireless connectivity, and its specifications may vary according to the model chosen. Model mANTBox 15s has an integrated 15 dBi sector antenna with 120-degree coverage. Its enclosure rating is IP54.

Purpose of the equipment: Such equipment will be used for connecting secondary repeater towers and distributing network traffic between backbone sites and nearby communities or public facilities.

Advantages: Sector antennas allow for directing wireless connectivity towards certain locations and thus allow better coverage and efficient use of wireless bandwidth.

The final choice of equipment should take into account the required coverage, wireless bandwidth, environmental factors, and power consumption. Installation of equipment should have appropriate mounting and protection against bad weather conditions.

1.1.4 Wi-Fi Access Points: Ubiquiti UniFi AC Mesh Pro (UAP-AC-M-PRO)

The Ubiquiti UniFi AC Mesh Pro is recommended for connecting the community via selected clinics, schools, community centers and public buildings using wireless networking technology.

Features: Dual-band Wi-Fi 5, 3 x 3 MIMO on both 2.4 GHz and 5 GHz bands, Gigabit Ethernet connectivity, IPX4 weather-resistant design.

Position in the network: These access points will enable users to connect their smartphones, laptops, tablets and other devices to the community network.

Advantages: Using centralised management will make it easier for IT specialists to configure the access points, monitor performance, handle guest access and address issues of connectivity.

Access points have to be placed according to the coverage needs of particular sites. Public Wi-Fi clients have to be isolated from critical municipal and healthcare systems using proper VLAN, firewall rules and client isolation capabilities.

1.2 Off-Grid Solar Power System (Per Remote Tower Node)

With the lack of stable electricity supply in rural areas, certain remote repeater towers will be supplied with solar energy systems that will decrease dependency on the electricity grid, ensuring continuity of communication in the event of load shedding.

1.2.1 Photovoltaic Solar System

For every remote tower, four 455 W monocrystalline solar panels have been allocated with a total installed capacity of 1,820 W.

These solar panels will be mounted on strong steel racks that are resistant to corrosion. The angle and direction of mounting will depend on location, shading and seasonality analysis of solar resources. North direction is recommended for installation in South Africa, although the suggested 32-degree angle should be confirmed at every site.

1.2.2 Solar Charge Controller

The Victron Energy SmartSolar MPPT 150/35 is an intelligent device that controls charging of the battery bank from the solar panels. It supports 12 V, 24 V, and 48 V battery systems, and has a maximum rated charging current of 35 A.

This controller is equipped with maximum power point tracking (MPPT), which helps with better exploitation of solar power. The voltage and current limits should be compared with the chosen solar panel system. Remote monitoring can be done by the use of appropriate monitoring equipment.

1.2.3 Battery Energy Storage System

For each remote tower, we will provide a 48 V, 200 Ah LiFePO4 battery with a built-in battery management system (BMS).

The nominal stored energy is: 48 V × 200 Ah = 9,600 Wh = 9.6 kWh.

Considering an 80% usable depth of discharge, the nominal usable energy will be around 7.68 kWh without considering any loss or aging of the battery.

72 hours of backup time is the objective and it should be verified with the calculation of power consumption. At 80% usable capacity, the battery could in theory sustain an average load of about 107 W for 72 hours without considering any losses. Therefore, the capacity of the battery must be defined based on the total consumption of the radios and switches etc.

1.2.4 Power Distribution Unit and PoE Switch

For the purpose of delivering network connection and power to the suitable devices, a Netonix WS-12-250-DC outdoor Gigabit PoE switch using DC power is recommended.

While DC-based power distribution reduces the amount of energy that would be wasted through the conversion of battery power into AC current and then again into DC current, the voltage levels and power restrictions need to be considered based on the needs of the connected devices. The passive PoE and active PoE standards should not be regarded interchangeably without the suitable equipment.

Suitable fuses and circuit breakers need to be used for the safety of the equipment.

1.2.5 Enclosure for Equipment and Lightning Protection

The equipment at each remote site requires an appropriate enclosure along with proper mounting, cable glands, bonding and grounding, and effective lightning and surge protection. It is necessary that the enclosure offers protection from dust, moisture, and heat.

Anti-theft mounting and restricted access to the equipment cabinet will help secure the infrastructure. Regular inspections can also be conducted to check any damage to cables, solar panels, batteries, and radios.

1.3 Hardware Requirements Specification Table

The below table gives an outline of the hardware requirement for Phase 1. These numbers are provisional and need to be verified by surveys, network design and power calculations.

Equipment category	Model/component	Quantity (Phase 1)	Key technical function
Core routers	MikroTik CCR2116-12G-4S+	2	Central routing, VLAN segmentation and network redundancy
Backhaul PtP radios	Ubiquiti airFiber 11	8 pairs	High-capacity wireless backbone links between selected sites
Sector/repeater radios	MikroTik NetMetal 5 / mANTBox	24 units	Wireless distribution to secondary towers and community sites
Wi-Fi access points	Ubiquiti UniFi AC Mesh Pro	60 units	Public Wi-Fi access at selected community facilities
Solar panels	Canadian Solar 455 W monocrystalline	64 panels	Renewable energy generation at 16 remote tower sites
Lithium batteries	Freedom Won eTower, 48 V, 200 Ah	16 units	Energy storage for remote tower equipment
DC PoE switches	Netonix WS-12-250-DC	16 units	DC power distribution and network switching
Enclosures and protection	Weather-rated cabinets, grounding and surge protection	16 sets	Environmental protection and electrical safety

Implementation issues: There are eight sets of radios that make a total of 16 radios. The recommended 64 solar panels will provide four solar panels per tower at 16 different locations, which means that there will be 1.82 kW installed capacity of solar power per tower. The last design should account for the access points' number and location, as well as the backhaul capacity and the equipment required for the connection of around 3,000 households.

2. SYSTEM AND APPLICATION SOFTWARE

There must be software that can be used in managing the network infrastructure and also protect any sensitive data and help monitor the performance of the network infrastructure and enable daily activities of the municipal workers, health practitioners, and citizens.

The suggested software will include the network operating system, server software, firewall and authentication service, monitoring and business applications. It should be compatible with the selected hardware.

2.1 System Software Needs and Solutions

System software	Usage in the Central Karoo Network	Improvement to the network solution
MikroTik RouterOS	Runs on compatible MikroTik routers and wireless devices to manage routing, interfaces, firewall rules and bandwidth.	Supports VLAN segmentation, routing between sites, traffic prioritisation and centralised network configuration.
Ubuntu Server LTS	Hosts selected server applications, databases and network services at the NOC.	Provides a stable, open-source server platform that can reduce software licensing costs and support a range of network services.
pfSense Firewall	Provides perimeter firewalling, traffic filtering and security controls at the NOC where a dedicated firewall is required.	Restricts unauthorised traffic and helps protect municipal systems from external threats.
FreeRADIUS	Provides authentication, authorisation and accounting (AAA) for supported network access services.	Supports centralised authentication and access policies for employees, administrators and external suppliers.
Zabbix Network Monitoring	Monitors network devices, servers, wireless links and compatible power-system telemetry.	Provides alerts and performance information to help technicians identify faults and perform proactive maintenance.
WireGuard or OpenVPN	Establishes encrypted VPN connections for authorised remote users and inter-site communication.	Protects data transmitted across untrusted networks and supports secure remote administration.

2.1.1 Network Operating Systems

MikroTik RouterOS will control compatible routing and wireless devices whereas Ubuntu Server LTS will serve as the operational system environment for chosen services running on the NOC.

Routing and VLAN protocols and firewall configurations will be used to segregate traffic of the city municipality, healthcare facilities, management and open public access. Dynamic routing and automatic failover will be set up when supported by the chosen equipment and appropriate for the network architecture.

OpenWrt can be deployed on compatible equipment where its capabilities meet the needs of the installation. However, OpenWrt and RouterOS cannot be considered as the same operating systems for all devices. (Gido et al., 2018).

2.1.2 Firewall and Network Security

The proposed firewall system pfSense will add an additional security layer between the internal network and external links. The firewall rules will limit access only to those services that are allowed and prevent any external network user to have access to sensitive government systems.

Suppliers need to be given individual accounts and access only to those systems necessary for their work. Other controls include authentication, multifactor authentication where possible, limited access through VPN, monitoring and regular review of the supplier's access. (Du, 2013; Shahjee & Ware, 2022)

This will help solve some of the issues related to lack of trust and unauthorised access to the government information.

2.1.3 Authentication and Access Control

FreeRADIUS will manage the authentication, authorization, and accounting of compatible network access services. Controlled access could be provided to municipal workers, authorized contractors, and other permitted personnel.

Permissions should be granted according to the role of each user. For instance, a third-party company whose job is to maintain the particular software should not have access to hospital systems or municipal accounts.

It is required that network segmentation and firewall rules accommodate such access restrictions. FreeRADIUS implements the authentication functionality; however, it does not block unauthorized network traffic on its own.

2.1.4 Network Monitoring and Maintenance

Monitoring of server availability, network equipment, wireless connections, interface usage and many more performance metrics will be carried out using Zabbix.

In situations where suitable monitoring devices are available, the network can monitor the voltage level, state of charge and temperature levels from remote power sources through suitable telemetry ports. (Tarigan & Mangani, 2018).

The alert system will assist technicians in detecting outages and link degradation among other anomalies.

2.1.5 Secure Remote Connectivity

The implementation of either WireGuard or OpenVPN will deliver encrypted VPN connectivity to authorised remote users and selected inter-site services. These connections will be able to facilitate remote network management and inter-facility communications.

The access to the VPN will need to be limited via individual accounts and permissions as well as firewall settings. The encryption serves the purpose of safeguarding data being transferred; however, the POPIA also needs the relevant data-handling policies to be put in place. (Gido et al., 2018).

2.2 Application Software Requirements and Solutions

The application software is going to facilitate the operation of the municipal offices in the district and its hospitals and residents. The software applications that will be chosen will have to be available through the network.

Application software	Usage in the Central Karoo Network	Improvement to the network solution
Municipal ERP system, such as SAP or a suitable municipal platform	Supports financial administration, procurement, reporting and other municipal processes for approximately 200 government employees across 28 offices.	Improves information sharing and supports more efficient administrative and financial processes.
OpenEMR and District Health Information System (DHIS2)	OpenEMR supports electronic clinical records, while DHIS2 supports health information collection, analysis and reporting.	Supports healthcare information management and reporting between participating facilities, subject to system integration and access permissions.
Microsoft 365 / Microsoft Office	Supports document preparation, spreadsheets, email and collaboration among municipal employees.	Improves communication and collaboration between departments and geographically separated offices.
Nextcloud	Provides centrally managed file storage, synchronisation and sharing when hosted on approved infrastructure.	Can maintain access to locally hosted files during external internet outages if the internal network and servers remain operational.
Zabbix or PRTG Network Monitor	Provides dashboards showing device availability, bandwidth utilisation and network performance.	Helps IT technicians identify network faults and monitor service performance.
Community e-Services Portal	Provides residents with access to selected municipal services, community information, educational resources and agricultural market information.	Improves access to digital services and supports better communication between residents and municipal authorities.

POLICIES AND STANDARD OPERATING PROCEDURES (SOPs)

3.1 Introduction and Governance Framework

Gido et al. (2018),the establishment of the Central Karoo District Community Network represents a critical infrastructure initiative aimed at delivering reliable, affordable digital connectivity across vast, geographically dispersed, and socio-economically vulnerable rural communities. To ensure the long-term operational sustainability, security, and resilience of this network, a robust framework of administrative policies and Standard Operating Procedures (SOPs) is essential. Rural telecommunications infrastructure faces unique operational challenges, including frequent load shedding and grid instability, extreme weather conditions, physical security threats (such as equipment theft and vandalism), and limited local technical support staff on-site.

This section provides comprehensive governance documentation tailored specifically to the operational context of the Central Karoo District. The policies and SOPs detailed below establish formal operational standards for business continuity, disaster recovery, network security, a acceptable usage, and ongoing physical and preventive maintenance.

3.2 Business Continuity Plan and Disaster Recovery Policy

3.2.1 Purpose and Scope

The purpose of the Business Continuity Plan and Disaster Recovery Policy is to establish operational protocols that guarantee business continuity and rapid disaster recovery for the Central Karoo District Community Network. This policy applies to all physical equipment, wireless backhaul links, central server environments, local solar power assemblies, and administrative data repositories maintained across the region.

3.2.2 Key Operational Metrics

To minimise disruption to municipal services, health clinics, schools, and community access points, the following recovery metrics are mandated:

Recovery Time Objective:

Core Network Infrastructure (Backhaul & Central Routers): RTO - 4 hours.

Community Access Nodes & Edge Wi-Fi Hotspots: RTO - 12 hours.

Non-Critical Administrative Support Systems: RTO - 24 hours.

Recovery Point Objective:

Network Configuration & Billing/Authentication Databases: RPO - 1 hour (achieved via real-time and hourly automated cloud backups).

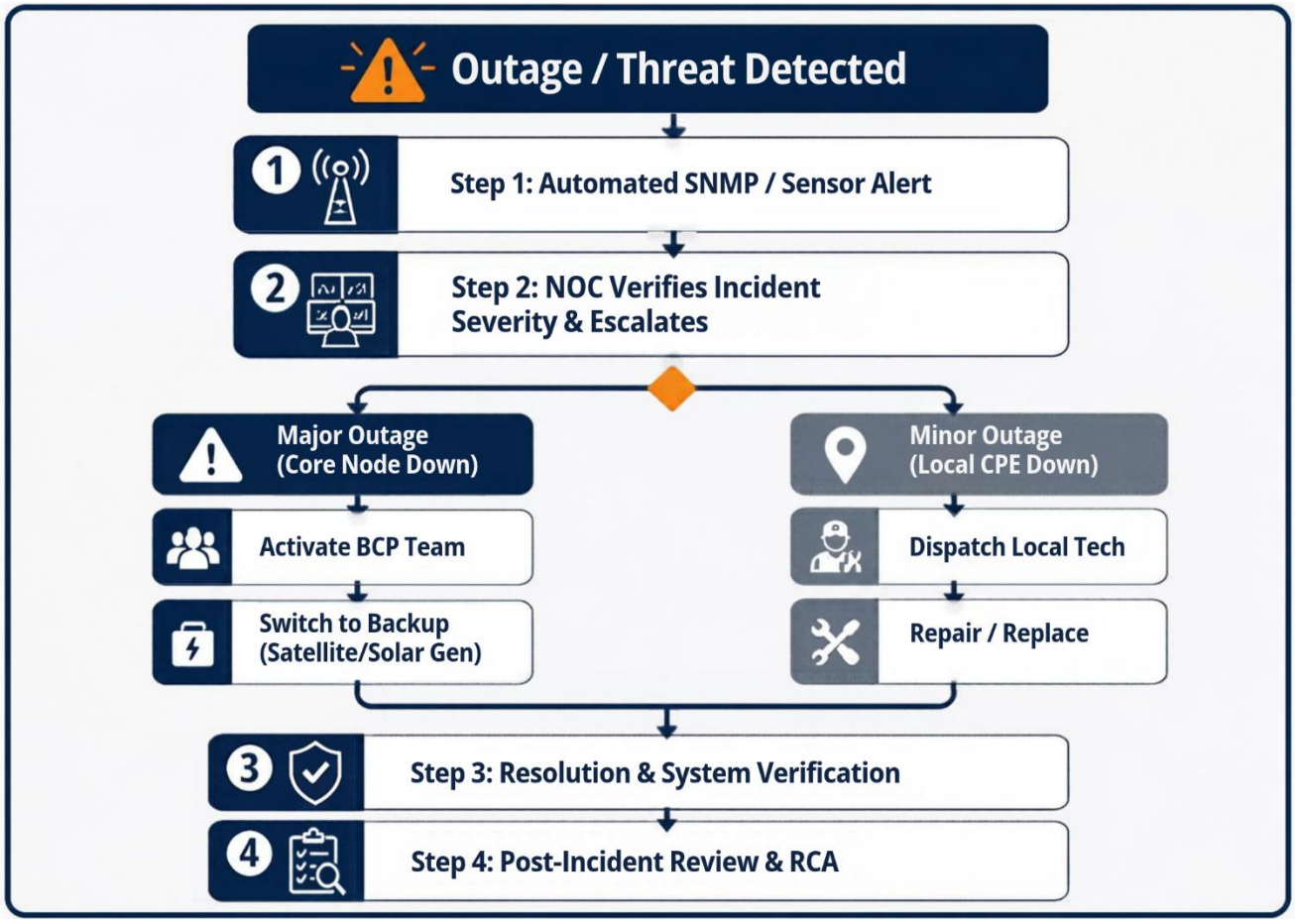
System Logs and Performance Analytics: RPO - 24 hours.

3.2.3 Business Impact Analysis (BIA) and Threat Assessment

Risk / Threat Vector	Severity	Impact Area	Mitigation & Continuity Strategy
Severe Load Shedding / Power Grid Collapse	High	High Tower & Backhaul Nodes	Off-grid hybrid solar-lithium battery backup system capable of sustaining operations for 72 hours without solar input. Automated SMS/SNMP alerts sent upon grid mains power loss.
Severe Weather (Lightning / Heatwaves)	Medium	High Towers & Outdoor CPE	Industrial-grade surge suppression, gas discharge tube lightning arresters, certified grounding 5Omega, and thermally regulated equipment enclosures.
Vandalism and Solar Panel Theft	High	High Tower Sites & Remote Repeaters	Anti-theft physical cages, tamper detection sensors, GPS tracker embedding inside battery banks, and community watch integration.
Upstream Fiber Cut / ISP Backhaul Outage	Critical	Entire District Connectivity	Dual-homed ISP failover (Primary: Optical Fiber Backhaul; Secondary: High-throughput Satellite link / LTE Failover via Mikrotik router scripts).

3.2.4 Incident Response and Disaster Recovery Procedure (SOP-BCP-01)

Objective: To provide step-by-step instructions for technical personnel during a critical network outage, extended power failure, or physical site damage.



Step-by-Step Execution:

Detection and Triage: Automated network monitoring tools (e.g., Zabbix / PRTG) monitor ping status, link throughput, and battery voltage levels every 60 seconds. Upon detecting a site node failure, an automated notification triggers an alert to the Network Operations Centre duty engineer.

1. Classification and Activation:

Level 1 (Minor): Single hotspot offline; low impact. Local technicians dispatched within 8 business hours.

Level 2 (Major): Repeater site offline, affecting a town or village. NOC engineer remotely routes traffic around the failure using mesh/dynamic routing protocols (OSPF). Field technician dispatched within 2 hours.

Level 3 (Critical): Core distribution router failure or total backhaul link disruption. BCP Team Lead declares formal BCP activation. Failover to satellite backhaul is initiated automatically or manually within 15 minutes.

2. Restoration and Site Recovery:

Technicians deployed to high towers must bring pre-configured spare units (Mikrotik/Ubiquiti swap units), replacement solar charge controllers, and portable fuel generators if battery reserves fall below 15%.

In the case of configuration corruption, restored settings are pushed directly from the secure cloud backup repo using Automated Ansible/RouterOS scripts

3. Post-Incident Review (Root Cause Analysis):

Within 48 hours of service restoration, the Technical Lead must submit a Root Cause Analysis (RCA) report outlining the outage cause, total downtime, financial impact, and preventative adjustments to procedures.

3.3 Network Security and Acceptable Use Policy

3.3.1 Security Policy Framework

To protect the integrity of the Central Karoo District Community Network, strict security controls must be enforced across all physical and logical entry points.

Network Segmentation:

Management VLAN (VLAN 10): Isolated network reserved exclusively for router, switch, and solar controller management. Accessible only via encrypted SSH/VPN tunnels.

Municipal / Healthcare VLAN (VLAN 20): High-priority traffic designated for municipal administration and rural clinic health data. Protected with WPA3-Enterprise authentication.

Public Hotspot VLAN (VLAN 30): Fully isolated user network with client isolation enabled (preventing communication between connected public devices).

Access Control and Authentication:

Public access is controlled via a centralised RADIUS captive portal requiring OTP registration.

Administrative access requires Multi-Factor Authentication and Role-Based Access Control.

3.3.2 Standard Operating Procedure: User Management and Access Control

Objective:

To standardise the provisioning, management, and de-provisioning of administrative access rights across network infrastructure components.

Procedure Steps:

1. Account Creation & Role Assignment:

Administrative accounts are created strictly upon formal authorisation from the IT Operations Manager.

Default admin credentials on all hardware (routers, switches, wireless access points) must be changed prior to field deployment.

Accounts are assigned to specific user groups based on the principle of least privilege:

Group 1 (Read-Only Technicians): Telemetry and log viewing only.

Group 2 (Field Support Technicians): Interface resets and basic wireless configuration.

Group 3 (Network Engineers): Full administrative access, routing updates, and firewall modification.

2. Credential Management and Storage:

All administrative credentials must be stored in an enterprise-grade password vault with strict access logging.

Passwords must be a minimum of 16 characters in length, combining uppercase, lowercase, numerical, and special characters.

Passwords must be rotated every 90 days or immediately following the resignation or termination of technical staff.

3. Deprovisioning:

Upon notification of employee offboarding, HR generates an immediate system alert.

System administrators must revoke all user accounts, active VPN tokens, and SSH keys within 1 hour of offboarding confirmation.

3.3.3 Acceptable Use Policy Overview (Public & Institutional Users)

All end-users connecting to the Central Karoo District Community Network must accept the AUP via the captive portal prior to accessing the internet.

Prohibited Activities:

Engaging in illegal downloading, peer-to-peer file sharing (BitTorrent), copyright infringement, or accessing unauthorised/unlawful content.

Launching Denial of Service attacks, network scanning, or attempting unauthorised penetration of public or private systems.

Hosting commercial web services or reselling community bandwidth without formal authorisation.

Bandwidth Management and Fair Usage Policy:

Rate-limiting per user (5 Mbps down / 2 Mbps up) enforced at the captive portal layer.

Daily data allocations (1 GB per user per day free tier) to ensure equitable distribution across community members.

3.4 Physical & Preventive Maintenance Standard Operating Procedures

3.4.1 Standard Operating Procedure: Solar Assembly and High-Tower Site Maintenance.

Objective:

To perform routine preventive maintenance on off-grid solar-powered wireless repeaters and high-tower installations across the Central Karoo District to minimise unscheduled system downtime.

Maintenance Checklist & Action Steps:

1. Safety Preparation:

Technicians must inspect high-altitude Safety Harnesses and Fall Protection Equipment prior to leaving the central depot.

Minimum two technicians required per site visit for high-tower work.

2. Solar and Battery Subsystem Check:

Solar Panels: Inspect photovoltaic panels for dust accumulation, bird droppings, or micro-cracks. Clean panel glass using demineralised water and soft brushes.

Charge Controller: Record Open Circuit Voltage, Short Circuit Current, and charge controller status logs. Ensure firmware is updated.

Lithium / Gel Battery Bank: Perform thermal camera sweep to identify hot spots in wiring terminals. Verify terminal torque settings (tighten loose connections to prevent arcing). Measure individual cell voltage levels.

3. Enclosure and Environmental Controls:

Replace dust filter pads on IP55/IP66 weatherproof outdoor cabinets.

Verify cabinet cooling fans turn on at designated thermal thresholds of 35.

Inspect cabinet seals for signs of moisture, insect intrusion, or dust penetration. Replace degraded silicone seals.

4. Wireless Hardware and Tower Structure:

Check guy wire tension and inspect tower structural bolts for rust or fatigue.

Inspect Ethernet cable runs (Cat6 outdoor shielded) for UV degradation, outer jacket cracking, or water ingress.

Inspect weatherproof vulcanising tape around N-type RF connectors and antenna feedlines. Re-wrap degraded seals immediately.

Measure Ground/Earth Resistance using an Earth Resistance Tester. Clean or enhance grounding rods if resistance exceeds threshold.

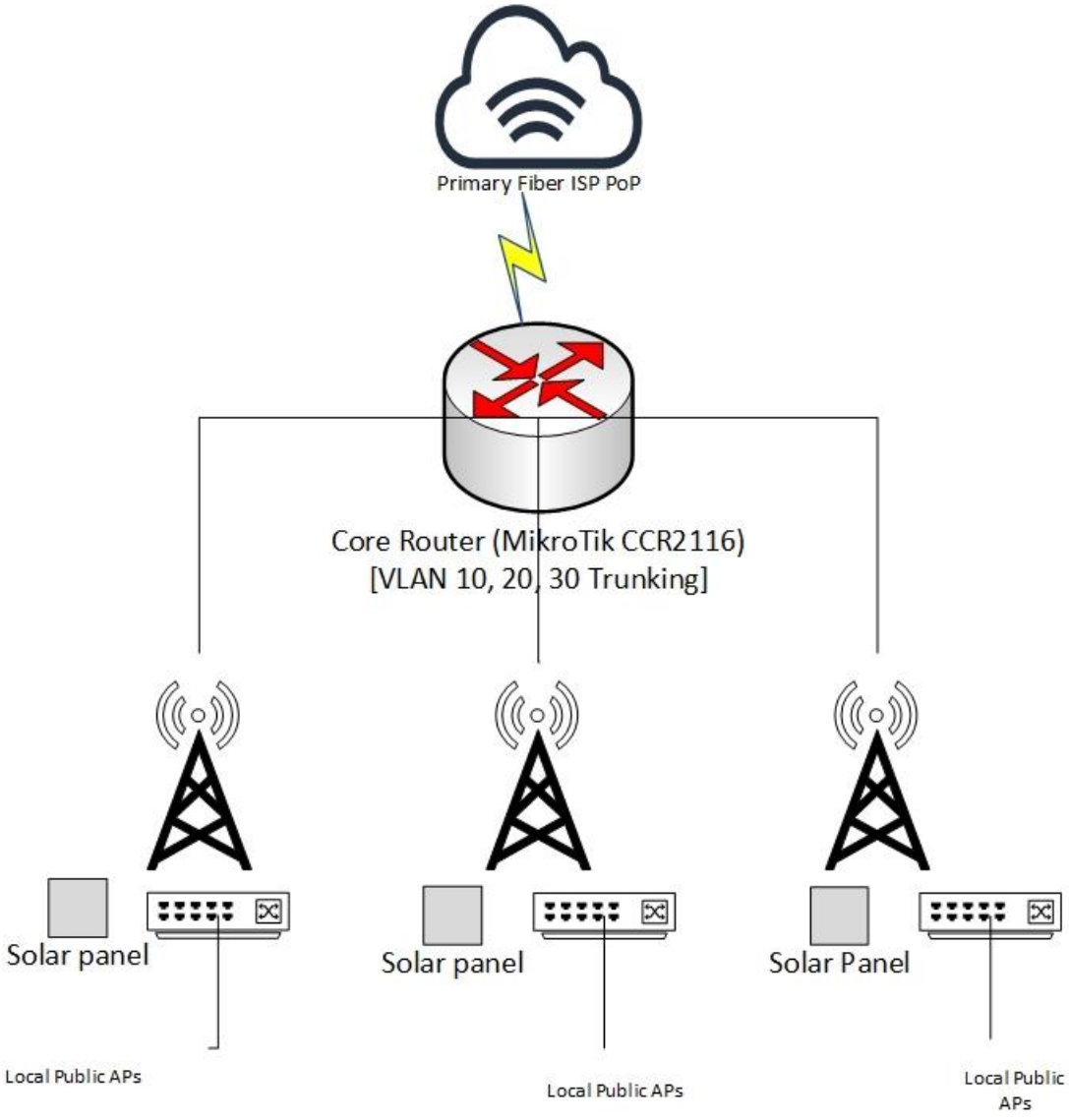
5. Completion and Sign-Off:

Submit digital site audit form including thermal images and before/after photos via the central asset management portal.

4. NETWORK SETUP (HARDWARE AND SOFTWARE INTEGRATION)

4.1 Physical & Logical Network Architecture

The Central Karoo District Network utilizes a Routed Mesh & Star Topology. A centralized core feeds main microwave links to local distribution towers, which fan out to public hotspots.



4.2 Network IP Addressing & VLAN Scheme

VLAN ID	Subnet CIDR	Target Assignment	Security / Control Rules
VLAN 10	10.100.0.0/20	Network Infrastructure Management	Isolated. SSH (Port 22), SNMP v3, HTTPS only via VPN.
VLAN 20	10.200.0.0/18	Municipal, Healthcare & Schools	Prioritized QoS. WPA3 Enterprise, inter-device traffic allowed.
VLAN 30	172.16.0.0/12	Public Wi-Fi Hotspots	Rate-limited via FreeRADIUS. Client isolation enforced.

5. DETAILED NETWORK ARCHITECTURE INTERACTION DIAGRAMS

The network architecture is organized across three main operational layers: Physical, Logical, and Cloud/Management.

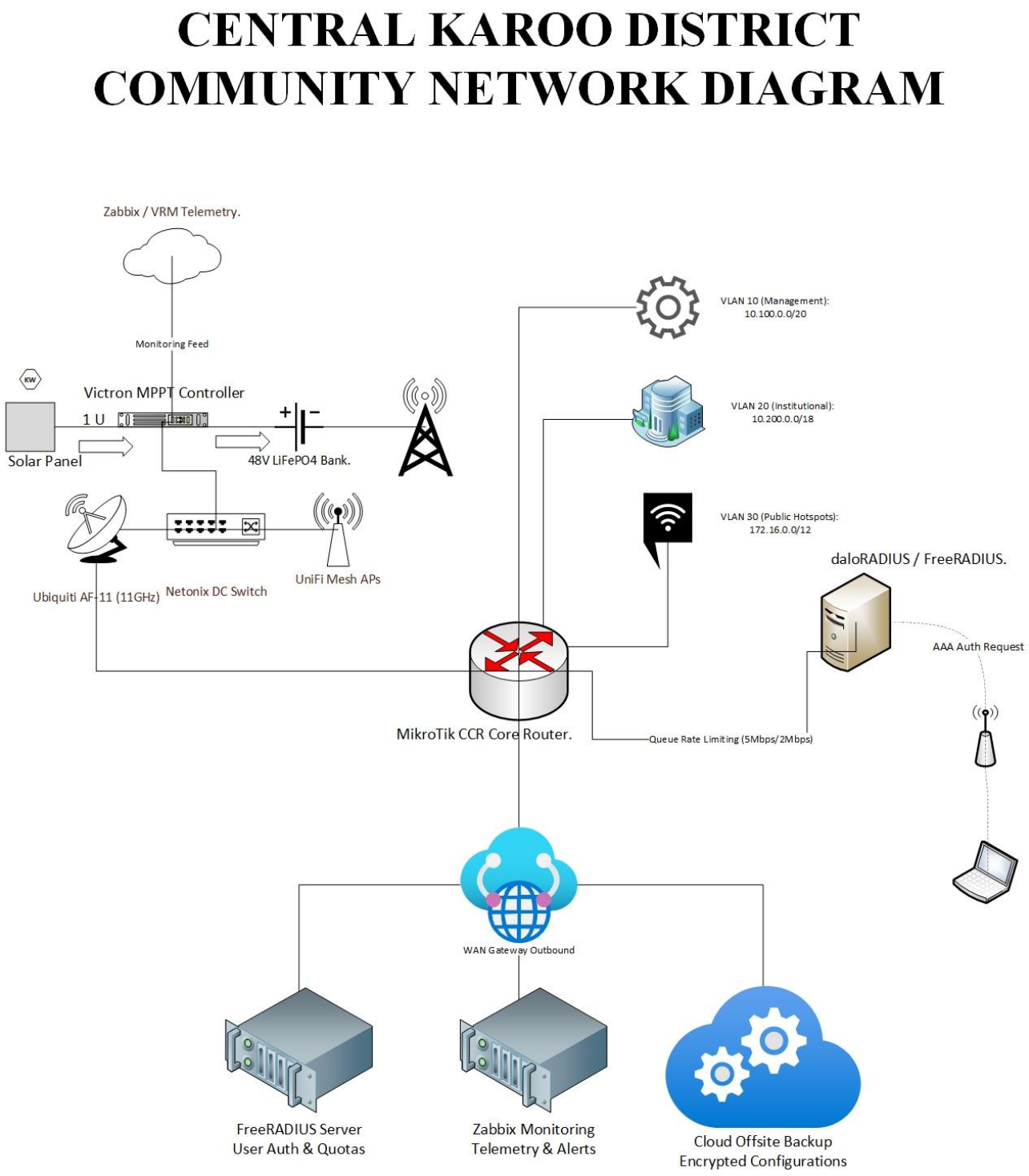


Figure Description: This diagram illustrates the high-level network topology for the Central Karoo District Community Network. It integrates a solar-powered off-grid backhaul and mesh Wi-Fi system, managed by a central MikroTik core router across three segmented VLANs (Management, Institutional, and Public Hotspots). The architecture incorporates AAA authentication with dynamic bandwidth rate limiting, continuous Zabbix telemetry monitoring, and secure cloud offsite configuration backups to ensure reliable community connectivity.

6. Outsourced Network Services

Some of the activities and operations of the network will be performed by specialised external companies, in order to enhance the performance of the network and ensure reliable services. It will enable the organisation to benefit from professional support and also save them the expenses of running all their services in-house.

6.1 Internet Connectivity & Fibre Services

Service Provider: A national fibre provider, like Liquid Intelligent Technologies or Openserve.

Services Provided: The provider will provide a dedicated, high-speed 1 Gbps Internet connection at the Beaufort West central hub. The internet connection will be engineered to achieve a consistent internet connection with a goal of 99.95% uptime as agreed in the service contract.

Technical Support: The provider will provide support services 24/7/365. For physical fibre faults, the target repair time will be 2 hours, depending on the contract agreed.

Because of the use of an external provider, you can be certain that the community will have a stable internet. It will also decrease the need for the organisation to maintain and repair the fibre infrastructure by itself.

6.2 Cloud Storage and Remote Server Management

Service Provider: Amazon Web Services running their Cape Town data centre region.

Services Provided: AWS will store and manage software and information to monitor and control network. It will also keep the network settings, system records and backups of important files safe.

Benefits: This will help minimize the need to rely on equipment in the Beaufort West central site. This is critical as it may lead to equipment damage or loss of service due to extreme weather and outages. If there is a failure or disaster, it will be easier to retrieve important data when there are secure backups offsite.

6.3 Specialist Tower Maintenance and Civil Works

Outsourced Vendor: Certified Regional Telecommunications Rigging Contractor.

Service Scope: High-tower climbing, annual structural guy-wire tension auditing, heavy antenna alignment, certified lightning protection system earth-ground installation, and physical tower structural repairs.

6.4 Managed Cybersecurity and Third-Party Access Monitoring

Outsourced Vendor: Accredited Managed Security Service Provider like Cybersoft, Dimension Data or Liquid C2.

Security Services: 24/7 Security Event Monitoring, Firewall & RADIUS Log Aggregation, Intrusion Detection Alerting, Quarterly Vulnerability Scanning and Annual Penetration Testing.

Reasoning: Addresses noted trust and security issues with external suppliers accessing government data. The district is unable to staff in-house to provide independent monitoring, and independent monitoring creates an auditable record of supplier activity, and aids compliance with POPIA.

6.5 Customer Support and Help Desk with both Tier 1 and 2

Outsourced Vendor: Third Party ICT managed helpdesk provider based in South Africa with language support.

Service Scope: Fault logging, password resets, Subscriber onboarding (for 3000 Houses), basic troubleshooting, Tier 3 escalation to District Network Engineers.

SLA Targets: 8 hours a day 7 days a week availability for residential customers with 24/7 availability for hospitals and government buildings; and first response within 30 minutes for priority customers.

6.6 Power Systems & Solar/Battery Maintenance

Outsourced Vendor: Solar PV and energy storage contractor who is recognised by SAPVIA.

Service Scope: 24-panel array inspection, cleaning, battery bank health check (70 x 200 Ah deep cycle units), inverter service, solar regulator check and end-of-life battery replacement twice a year.

Rationale: Power continuity at the NOC is the backbone of the entire network system and battery degradation is the most common issue that leads to failures at rural sites. Specialist testing causes reduced battery life and can prevent outages.

6.7 Managed Wi-Fi and Last-Mile Customer Installation

Outsourced Vendor: Local certified ISP installation contractors who are partners in running small businesses in the district.

Service Scope: The services to be rendered are installation of Customer Premises Equipment at 3000 homes, testing of their services, antenna alignment and replacement of faulty units.

Benefits: Delivers local jobs and skills transfer, and the installation process can be done in one year without the need to have to wait for one installation crew to finish while the other is working.

6.8 Hardware Supply, Warranty and Spares Logistics.

Outsourced Vendor: Company that is an authorised distributor/reseller of the chosen wireless platform such as Smart Bridges, Ubiquiti, MikroTik or Cambium partners.

Service Scope: Provide access points, client units and switches, 3-year extended warranties, advance replacement RMA and critical spares stocked in Beaufort West.

Rationale: There is a long-time lag for rural parts to get replaced from a metro centre. MTTR decreased using local spares and advance replacement.

6.9 Regulatory, Licensing and Compliance Advisory

Outsourced Vendor: Telecommunications regulatory consultant or firm.

Services Scope: Annual compliance review, ICASA licensing, spectrum compliance for the point-to-multipoint links & POPIA data protection impact assessment.

6.10 Hospital Systems and ERP Support (Application-Level)

Outsourced Vendor: Vendor's certified support partner.

Service Scope: Application patching, ERP access review for users and secure VPN onboarding for authorised external suppliers.

Rationale: It will isolate network operations from application support and regulate and track access by suppliers.

7. OTHER IT SUPPORT FACILITATING BUSINESS PROCESSES

To streamline operations and support community stakeholders, the network incorporates specialized IT tools and business process automation. The technology should be able to support the main processes of the district (which are healthcare, administrative and community access) thanks to an IT alignment with the milestones found throughout the project.



1. IT Support for Milestones 1-3 (Infrastructure and Hospital Integration)

Database Administration Support: During the crucial switch of month 4, the specialized approach of the database administrators is needed for the transferring of the patient data from the ISP into another dedicated network. This support provides the fact that during the treatment process in the 40 hospital blocks, no information will be lost.

Network Architect Support: Adequate engineering support is needed for the installation of main routers and realization of the VLAN segmentation described in the logical diagrams so that the ERP system of the 200 state employees will be secure.

2. IT Support for Milestones 4-5 (Residential Rollout)

Field Installation Teams: Special technical teams are sent all over to install special weatherproof CPE (Customer Premises Equipment) in the 3000 residential houses.

Community Helpdesk: Local IT helpdesk has been organizing in order to help the local people with any troubleshooting issues that may happen while using their internet connection.

3. IT Support for Milestone 6 (Training and Handover)

NOC Operations Training: IT professionals are brought in to execute a training program to develop local municipal employees' skills. They give on-site guidance for the operations of the network diagrams, both theoretical and physical, as well as for the functioning of the solar and battery systems.

VIP Application Support: Comprehensive desktop support is provided to government representatives regarding the connection problems they might be experiencing with Microsoft Office or the ERP application during this final transition stage (Bahr & Bahr, 2020; Gartner, 2022).

Reference list:

- Gido, J., Clements, J., & Baker, R. (2018). Successful project management (7th ed.). Cengage Learning.
- Gartner. (2022). Toolkit: IT hardware evaluation matrix and vendor proposal scoring model. Gartner Research. Available at: https://www.gartner.com [Accessed on 14 October 2026].